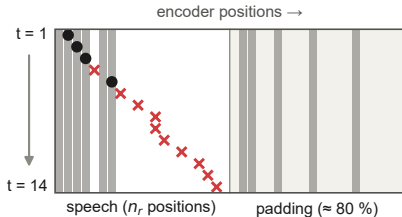


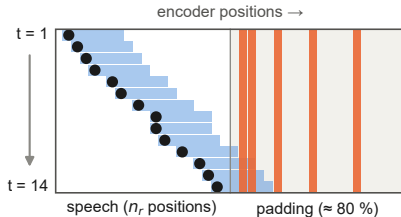
(a) One-shot retention

H2O, SnapKV, PyramidKV, sink + one-shot audio



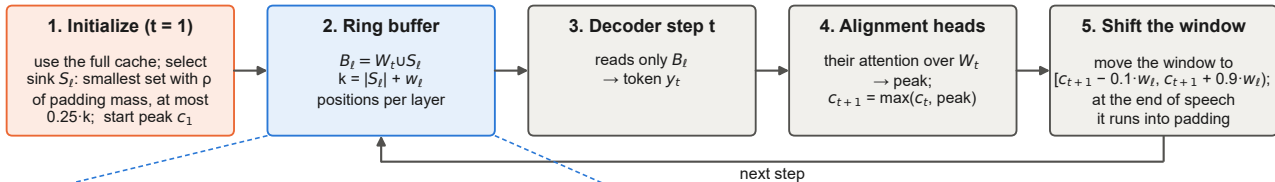
(b) PadSink-Track

the selection is renewed at every step



- position the decoder needs at step t
- ✗ needed but evicted → errors, loops
- one-shot set: chosen at $t = 1$, fixed
- window W_t : follows the alignment peak c_t
- padding sink S_t : exact K/V, chosen at $t = 1$

(c) One decoding step of PadSink-Track



Ring buffer of one layer (size k)



Slots of positions that left the window (101, 102) are overwritten with the entering ones (109, 110), with no per-step gather. s = sink slots. Attention does not depend on slot order.